

PIRVISION_FOG_ PRESENCE_DETECTION

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INTRODUCTION

- Fog can affect visibility and environmental conditions.
- Detecting fog conditions automatically can help in monitoring and safety applications.
- This project uses PIR sensor readings and temperature data.
- Machine Learning techniques are applied to identify patterns in the sensor data.
- The system aims to classify the given sensor observations based on the available Label.

PROBLEM STATEMENT AND OBJECTIVES

Problem:

Traditional environmental monitoring may require continuous observation and manual analysis.

Proposed Solution:

- Collect PIR sensor and temperature readings.
- Preprocess the collected dataset.
- Handle class imbalance using SMOTE.
- Select the most relevant features.
- Train different Machine Learning classification models.
- Compare the model performance using evaluation metrics.

The main objectives of this project are:

1. To analyze PIR sensor-based environmental data.
2. To identify useful features from the dataset.
3. To preprocess and clean the data.
4. To handle imbalanced classes using SMOTE.
5. To reduce the number of features using feature selection.
6. To train classification models.
7. To evaluate and compare the performance of different models.

DATASET DESCRIPTION

The dataset contains:

- 7,651 records
- 59 columns
- Date
- Time
- Temperature
- PIR sensor readings
- Label
- Dataset

<https://archive.ics.uci.edu/dataset/1101/pirvisi>
on_fog_presence_detection

The sensor features include:

- Temperature_F
- PIR_1 to PIR_55

Target Variable:

- Label
- Dataset size: 7651 rows $\times$ 59 column

DATA PREPROCESSING

The following preprocessing steps were performed:

- Dataset imported using Pandas.
- Data converted into a DataFrame.
- Dataset structure and values were examined.
- Date and Time were excluded from the Machine Learning input.
- Label values were processed.
- Outliers were handled using the IQR method.
- The cleaned data was prepared for further processing.
- Input features: Temperature + PIR sensor readings
- Target: Label

CLASS IMBALANCE AND SMOTE

The original data contained an imbalance in the target classes.

To solve this problem, SMOTE (Synthetic Minority Over-sampling Technique) was used.

SMOTE Process:

- Identifies minority-class samples.
- Generates synthetic samples.
- Balances the class distribution.
- Helps the model learn minority classes more effectively.

After SMOTE:

Total samples = 18,741

Each available class was balanced to:

6,247 samples per class

FEATURE SELECTION

Some selected important features include:

- Temperature_F
- PIR_2
- PIR_3
- PIR_4
- PIR_5
- PIR_9
- PIR_10
- PIR_11
- PIR_12
- PIR_13
- PIR_39
- PIR_40
- PIR_41
- PIR_42
- PIR_44
- PIR_45
- PIR_46
- PIR_47
- PIR_48
- PIR_49
- PIR_50
- PIR_51
- PIR_52

The project uses SelectKBest with ANOVA F-test (f_classif).

- Initially, the dataset contains 55 PIR sensor features + temperature.
- SelectKBest was used to select the top 25 features.
- This reduces unnecessary features.
- It can improve model efficiency and reduce computational complexity.

TRAIN AND TEST SPLIT

The prepared dataset was divided into:

- **Training Data – 80%**
- **Testing Data – 20%**

`train_test_split()` was used with:

`test_size = 0.2`

`random_state = 40`

Purpose:

- **Training data:** Used to train the models.
- **Testing data:** Used to evaluate how well the trained models perform on unseen data.

MACHINE LEARNING MODELS

Five classification algorithms were used:

- Logistic Regression
- Decision Tree
- Random Forest
- AdaBoost
- Gradient Boosting

These models were trained using the selected and scaled features.

FINAL OUTPUT

	Model	accuracy	precision	f1	recall
0	Logistic Regression	0.99	0.99	0.99	0.99
1	Decision Tree	1.00	1.00	1.00	1.00
2	Random Forest	1.00	1.00	1.00	1.00
3	AdaBoost	1.00	1.00	1.00	1.00
4	Gradient Boosting	1.00	1.00	1.00	1.00

CONCLUSION

- PIR sensor and temperature data can be processed using Machine Learning techniques.
- Data preprocessing improves the quality of the dataset.
- SMOTE helps to balance the classes.
- SelectKBest reduces the feature set from the original sensor variables to 25 important features.
- Multiple classification algorithms can be compared to identify a suitable model.
- The project provides a Machine Learning-based approach for analyzing PIR sensor observations related to fog presence detection.



**THANK
YOU**